

EXP 8: Implementation and Comparative Analysis of SARSA and Q-Learning

Aim

To implement **SARSA** and **Q-Learning** algorithms and compare their performance in learning the optimal policy in a Reinforcement Learning environment.

Algorithm

SARSA Algorithm

1. Initialize the Q-table .
2. Select an action using an ϵ -greedy policy.
3. Execute the action and observe reward and next state .
4. Select the next action using the current policy.
5. Update the Q-value:
6. Repeat until the episode ends.

Q-Learning Algorithm

1. Initialize the Q-table .
2. Select an action using an ϵ -greedy policy.
3. Execute the action and observe reward and next state .
4. Find the maximum Q-value of the next state.
5. Update the Q-value:
6. Repeat until the episode ends.

CODE

```
import numpy as np
import matplotlib.pyplot as plt

states = [0, 1, 2, 3]
actions = [0, 1]
rewards = {
    (0, 0): 0, (0, 1): 0,
    (1, 0): 0, (1, 1): 1,
    (2, 0): 0, (2, 1): 2
}
```

```

next_state = {
    (0, 0): 1, (0, 1): 2,
    (1, 0): 0, (1, 1): 3,
    (2, 0): 0, (2, 1): 3
}

gamma = 0.9
V = np.zeros(4)
for _ in range(100):
    new_V = V.copy()
    for s in states[:-1]:
        values = []
        for a in actions:
            ns = next_state[(s, a)]
            r = rewards[(s, a)]
            values.append(r + gamma * V[ns])
        new_V[s] = max(values)
    V = new_V
print("State Values:")
for s in states:
    print("State", s, ":", round(V[s], 3))
plt.figure(figsize=(7, 5))
plt.bar(states, V)
plt.xlabel("States")
plt.ylabel("Value")
plt.title("MDP State Value Function")
plt.xticks(states)
plt.grid(axis="y")
plt.show()

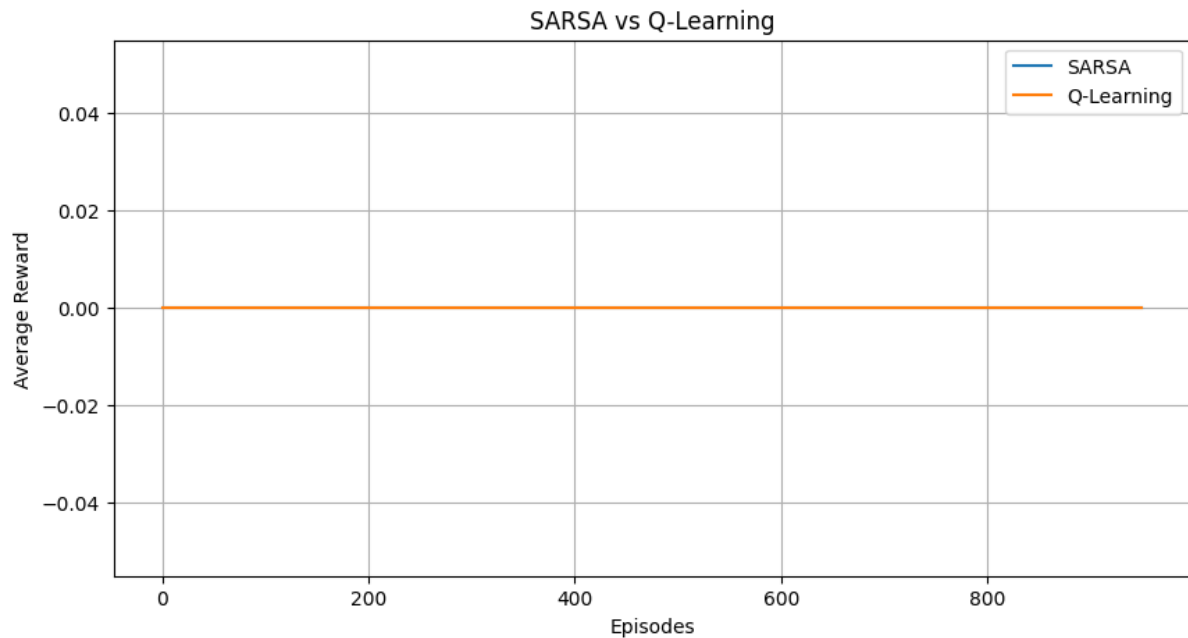
```

OUTPUT

Average Reward (Last 100 Episodes)

SARSA : 0.0

Q-Learning : 0.0



Result

Thus, **SARSA and Q-Learning** were successfully implemented and compared. SARSA learns using the action actually taken and follows an on-policy approach, whereas Q-Learning learns using the best possible action and follows an off-policy approach. Q-Learning generally converges faster to the optimal policy, while SARSA tends to learn safer policies.